

mrm in Practice: multilevel reconciliation methodology on ontario carceral, Toronto Police, and JuV data

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abstract

This paper is the empirical application companion to the mrm (multilevel reconciliation methodology) framework ([ruhela 2024](#)) and its open-source software realisation in the morie toolkit (Python and R [ruhela 2024a,b](#)). Two anchor datasets are analysed under the mrm modules: the Ontario Tracking Information System and restrictive-confinement detailed dataset (OITS and RICS, fiscal years 2013–2023, $n = 76,934$ person-day records over 65,467 unique person-year identifiers; the OITS and RICS segregation-placement file, $n = 82,001$ placements, is used for the per-placement analyses of §4.1), and the Toronto Police Service open-data major-crime corpus (TPJ). On OITS and RICS, a double-machine-learning interactive regression model (IRM-DM) estimates the effect of alert complexity (treatment $ac \geq 2$) on placement volatility (count of regional transfers per person-year vm), pooled across 2013–2023 and by fiscal year. The pooled estimate is $\hat{\tau}_{ATE} = 0.1605$ ($SE = 0.00628$, 95% CI $[0.1481, 0.1728]$), and the per-year breakdown shows a monotone rising trend ($\hat{\tau}_{ATE,2023} = 0.1342$, $\hat{\tau}_{ATE,2024} = 0.1591$, $\hat{\tau}_{ATE,2025} = 0.1737$). Multi-way regional clustering shifts the point estimate into a tighter band $\hat{\tau} \in [0.1932, 0.2013]$ across 36 region $\times$ scheme combinations, with all $p < 10^{-24}$. Matched-sample Poisson gamma (matchit::nearest-neighbour propensity-score matching followed by a random-intercept Poisson generalised linear mixed model) reproduces the direction of the effect as a sensitivity check. The empirical identifiability of cross-year recidivism on OITS is confirmed impossible by data-dictionary inspection (0/65,467 identifiers appear in two or more fiscal years). On TPJ crime events, a non-Markovian Hawkes process specification (gamma excitation kernel, sinusoidal baseline) is preferred over the classical Markovian Hawkes by Bayesian-information-criterion ranking and a higher Kolmogorov–Smirnov p -value on the rescaled inter-event times. Every numerical claim in this paper is reproducible from the morie toolkit through the code blocks in appendix A.

keywords: causal inference, double machine learning, observational inference, Ontario restrictive confinement, Toronto Police Service, Hawkes process, mrm framework, sociolegal statistics, R, Python.

i. introduction

This paper is the empirical companion to the mrm (multilevel reconciliation methodology) framework ([ruehela 2020d](#)) and the morie toolkit ([ruehela 2020c,b](#)). The mrm framework specifies a ten-estimator per-individual ensemble, an aggregate χ^2 family, and a self-exciting fiawkes specification (the methodology is in [ruehela 2020a](#)). The morie toolkit makes those modules available in r and Python. The present paper exercises the toolkit on two open canadian sources: the ontario tracking information system bui segregation detailed dataset ([2017 bui](#)) and the toronto police service major-crime open-data corpus ([2017](#)).

Three results are reported. First, an mrm- δ estimate of the effect of alert complexity on placement volatility on [2017 bui](#) over fiscal years 2013--2016 with multi-way regional clustering and matched-sample sensitivity (section [2](#)). Second, eleven statistical-physics-of-crime and institutional-churn result sections on [2017](#) and [2017 bui](#) (section [3](#)), covering per-category fiawkes fits (markovian and non-markovian), Lévy-flight mobility, urban-scaling exponents, spatial autocorrelation, inspection-game phase diagrams, reaction-diffusion, bulldorff space-time scan, yoffmanian institutional churn, stochastic forecasting, the crime severity index robustness check, and extended spatial diagnostics.

every numerical claim in this paper is reproducible from the morie toolkit. section [4](#) catalogues the morie calls that generate every table; the full code is in appendix [a](#). The intent is that a reader who installs morie at `pip install morie` or `install.packages('morie')` can recompute every reported estimate from the same open data sources.

The rest of the paper is organised as follows. section [5](#) documents the two data sources and the data-dictionary constraints under which the analysis is valid. section [6](#) states the identifying assumptions and the estimators. section [7](#) reports the population descriptives on [2017 bui](#). section [8](#) presents the mrm- δ estimates with their multi-way clustering robustness band and the matched-sample Poisson χ^2 sensitivity. section [9](#) reports the eleven statistical-physics-of-crime and institutional-churn result sections. section [4](#) catalogues the morie calls. section [10](#) states the limitations. section [11](#) discusses the substantive interpretation. section [12](#) concludes.

A. Data

Two open canadian sources are integrated under the mrm data-source notation of [ruehela \(2020d\)](#): [2017](#) (provincial restrictive-confinement microdata) and [2017](#) (municipal-police event-level open data).

A.1. ontario tracking information system ([2017 bui](#))

The [2017 bui](#) file is the ontario ministry of community safety and correctional services segregation detailed dataset, published annually on [data.ontario.ca](#) under the [bui](#) release of the restrictive-confinement open-data series. one row of [bui](#) represents a single segregation event for a single individual in a single restrictive-confinement placement.

after ingestion through `morie::mrm_otis_load()`, the [bui](#) restrictive-confinement corpus over fiscal years 2013, 2014 and 2016 has $n = 76,934$ row-level records covering 65,467 unique (identifier, fiscal-year) pairs. The companion [bui](#) segregation-placement file ($n = 82,001$ placements over the same fiscal years) is used in §[3.4](#) for the per-placement cramér's V and region-locality analyses.

Table 1: `bu` schema (verified against the official ontario `au`-`rcdd` data dictionary, november 2022 release: the dictionary has a known label-shift bug in rows 31--34, which is benign --- the column names in our schema are correct).

column	type	substantive role
<code>endfiscyear</code>	integer	fiscal year of placement end (2013--2022)
<code>uniqueindividual_id</code>	string	per-year-reset random identifier
<code>gender</code>	categorical	demographic stratifier
<code>region_attimeofplacement</code>	categorical	4 regions: Toronto, Eastern, Western, Central, Northern
<code>region_mostrecentplacement</code>	categorical	same domain
<code>age_category</code>	categorical	3 age bins
<code>numberconsecutive_days_segregation</code>	integer	event duration in days
<code>segreason_*</code> (8 cols)	binary	reason: security, protection, disciplinary, search, other
<code>mentalhealth_alert</code>	binary	alert indicator
<code>suiciderisk_alert</code>	binary	alert indicator
<code>suicidewatch_alert</code>	binary	alert indicator
<code>number_of_placements</code>	integer	total restrictive-confinement placements (within year)

A.1. identifier semantics on `bu`: per-year reset

The official data-dictionary entry for `uniqueindividual_id` specifies the format `YYYY-XXXXXX-file-tag`: `YYYY` is the fiscal-year prefix, `XXXXXX` a randomly-assigned integer, and `file-tag` a per-file suffix (`-rc` for `au` restrictive-confinement, `-sy` for `bu` segregation, `-dc` for `du` deaths in custody). The year prefix and file-tag suffix are baked into the identifier by design, so the identifier is randomly re-assigned to different individuals each year and is encoded distinctly per file even within the same year. Two consequences follow.

cross-year recidivism on `bu` is unanswerable. The empirical confirmation is that 0 of 65,467 identifiers appear in two or more fiscal years (this can be reproduced through `morie::mrm_otis_check_id_reset()`).

cross-file joining on identifier is invalid within the same fiscal year. joining `bu` to `au` or to any of the `c`-tables on `uniqueindividual_id` would semi-randomly pair distinct people. The `morie bu` pipeline therefore operates on `bu` alone for the per-individual analysis of Section 3.

A.3. Toronto Police Service (TPS) major-crime open data

TPS publishes daily incident-level records of major-crime occurrences under its open-data portal. Each row carries an event timestamp at day-level granularity, a major-crime category (assault, auto theft, break and enter, robbery, theft over), and spatial coordinates aggregated to neighbourhood. The `morie` ingestion routine `morie::mrm_tps_load()` produces a tidy event sequence $\{(t_i, c_i, \mathbf{x}_i)\}_{i=1}^N$ where $t_i \in [0, T]$ is the event time, c_i is the category, and $\mathbf{x}_i \in \mathbb{R}^2$ is the spatial coordinate.

3. methods

3.1. estimand on which build

The alert-complexity (ac) and placement-volatility (vm) features are person-year-level statistics defined on the sequence of placements within the fiscal year. They are the empirical contributions of the nmrj framework to the literature, distinct from per-placement metrics used in the prior solitary-confinement literature.

alert complexity (ac) — state diversity across placements. The three binary alerts (mental-health, suicide-watch, suicide-risk) partition every placement into $2^3 = 8$ mutually-exclusive status combinations, labelled $a_1, a_2, \dots, a_8$ as in Table A. At the person-year level, $a_{k,iy}$ denotes the count of placements of individual i in fiscal year y that fall in status combo k . The alert-complexity statistic is

$$ac_{iy} = \sum_{k \in \{1,2,4,5,7,8\}} \mathbf{1}\{a_{k,iy} > 0\}, \quad (\text{i})$$

i.e., the number of distinct alert-combo states (out of the six selected by the framework) in which the individual had any placement during the year. The framework selects six of the eight combinations: combinations a_3 (suicide-risk only) and a_6 (mental-health plus suicide-risk, no watch) are excluded.

Table A: The eight alert-state combinations partitioning placements by the three-alert profile (mental, watch, risk). The nmrj alert-complexity statistic ac in (i) counts the distinct states from $\{a_1, a_2, a_4, a_5, a_7, a_8\}$ that the individual had any placement in during the fiscal year.

Label	mental	watch	risk	in ac?
a_1	i	u	u	yes
a_2	u	i	u	yes
a_3	u	u	i	no
a_4	i	i	u	yes
a_5	u	i	i	yes
a_6	i	u	i	no
a_7	i	i	i	yes
a_8	u	u	u	yes

The binary treatment indicator is the high-alert-complexity indicator $D_{iy} = \mathbf{1}\{ac_{iy} \geq 2\}$ --- individuals whose within-year placement sequence covered at least two distinct alert-combo states (drawn from the six framework-selected states).

Placement volatility (vm) — regional-transition count. Let $\rho_{iy,t}$ denote the placement region for individual i in fiscal year y at placement index t (ordered chronologically within the year), and let $\rho'_{iy,t}$ denote the most-recent-placement region at the same row. The placement-volatility statistic is

$$vm_{iy} = \sum_t \mathbf{1}\left\{\rho_{iy,t} \neq \rho'_{iy,t} \text{ or } \rho_{iy,t} \neq \rho_{iy,t-1} \text{ with } \rho_{iy,t-1} \text{ defined}\right\}. \quad (\text{A})$$

This counts the number of regional transitions that individual i experiences across their within-year placement sequence. It is not a binary indicator of 'moved at all': vm_{iy} ranges

over the non-negative integers and is the natural count outcome on which the Poisson generalised-linear-mixed-model sensitivity check of section 3.2 operates.

The OTE estimand τ is the average treatment effect of D_{iy} on vm_{iy} :

$$\tau_{ATE} = \mathbb{E}[vm(1) - vm(0)], \quad \tau_{ATTE} = \mathbb{E}[vm(1) - vm(0) \mid D = 1]. \quad (3)$$

Covariates entering the OTE nuisance models are $X_{iy} = (\text{age_category}_{iy}, \text{gender}_{iy}, \text{region_at_time_of_placement}_{iy}, y)$. The treatment is $D_{iy} = \mathbf{1}\{\text{ac}_{iy} \geq 2\}$. The outcome is $Y_{iy} = vm_{iy}$.

3.1. Identifying assumptions

The identification rests on three standard assumptions:

- i. conditional exchangeability: $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$. reasonable when X captures the demographic and regional structure that drives both the alert-flag pattern and the regional-transfer logistics.
- ii. Positivity: $0 < \Pr(D = 1 \mid X) < 1$ for every level of X realised in the sample. empirically verified (section 2).
- iii. SUTVA: no interference across individuals and a single well-defined version of treatment. defensible at the placement-event level.

under (i)–(iii) the ATE and ATTE are identified by the standard OTE score function (Chernozhukov et al. 2018).

3.2. Primary estimator: IRM-OTE

The interactive regression model of Chernozhukov et al. (2018) is estimated through morie’s wrapper around doubleML (Bach et al. 2014, 2015). Nuisance functions $\hat{g}(x) = \mathbb{E}[Y \mid X = x, D = d]$ and $\hat{m}(x) = \Pr(D = 1 \mid X = x)$ are estimated with the default morie learner stack (regr.lm + classif.log_reg in the mlr3 family), $K = 5$ -fold sample splitting, $R = 2$ repetitions, and Neyman-orthogonal scoring.

3.3. Multi-way clustering schemes

Five regional clusters (Toronto, eastern, western, central, northern) and three clustering schemes are reported in section 2.3: (i) no clustering (the i.i.d. baseline), (ii) cluster on identifier alone, (iii) cluster on identifier and region jointly. A year-clustered specification (cluster on endfiscyear) and a year-with-identifier specification (uniqueindividual_id included as a covariate and clustered on year) are reported as additional robustness checks in appendix B.

3.4. Matched-sample Poisson GLMM sensitivity

A 1:1 nearest-neighbour propensity-score matching (matchrit with distance = glm($D \sim \text{age} + \text{gender} + \text{year}$)) constructs a covariate-balanced subsample. On the matched data a Poisson generalised linear mixed model $vm \sim D + \text{age} + \text{gender} + \text{year} + (1 \mid \text{region})$

recovers an incidence-rate-ratio interpretation of the treatment effect under a parametric Poisson distributional assumption. The agreement (or disagreement) between the mrnj- δ mrj estimate of section 3.3 and the γ lmj rate-ratio estimate is the central sensitivity check.

3.D. Hawkes specification on θ_j^*

The Hawkes self-exciting point-process specification of [Ruhela \(2018a\)](#) is fit to each θ_j^* major-crime category under four excitation kernels (exponential, gamma, Weibull, power-law/ λ omax) and two baseline parameterisations (constant, sinusoidal). Maximum-likelihood estimation, Bayesian-information-criterion ranking, and the time-rescaling-theorem goodness-of-fit ([Brown et al. 2014](#)) are computed by `morie::mrm_hawkes_tps()`.

4. Descriptive statistics

The pre-causal descriptive structure of $\mathcal{O}^*$ b μ i is summarised below: full breakdowns by age, gender, and region are produced by `morie::otis_descriptives()` and reproduced by the code block in section 6.

Table 3: Sample structure of $\mathcal{O}^*$ b μ i over fiscal 2013–2016 (verified ground-truth counts from the published $\mathcal{O}^*$ -rc analysis; reproducible from `morie::mrm_otis_load()`).

Quantity	value
Total row-level records (n)	76,934
unique person-year identifiers	65,467
fiscal years covered	2013, 2014, 2015
identifiers appearing in ≥ 2 fiscal years	0
regions of placement	Toronto, eastern, western, central, northern

The 0/65,467 figure confirms the data-dictionary statement that the identifier is re-assigned each fiscal year. The denominator 65,467 is the unique (identifier, year) cardinality and is treated as the per-individual sample size in sections 5 and 7.

5. Causal estimates: alert complexity on placement volatility

5.1. Pooled 2013–2016 δ mrj estimates

Under the mrnj- δ mrj specification of section 3.3, the pooled treatment effect of $D = \mathbf{1}\{\text{ac} \geq 2\}$ on $Y = \text{vm}$ over fiscal 2013–2016 is reported in Table 4.

Both estimands are statistically distinguishable from zero at every conventional level: the t-statistics correspond to $p < 10^{-143}$ for the $\alpha 0\epsilon$ and $p < 10^{-145}$ for the $\alpha 00\epsilon$. The point estimates correspond to a 16 percentage-point increase in the probability that the most recent placement is in a different region than the segregation-event placement, conditional on the alert-complexity indicator switching from low to high.

Table 4: Pooled 1993–1998 iron-δm estimates of τ_{ATE} and τ_{ATTE} for the treatment $D = 1\{\text{ac} \geq 2\}$ on the outcome $Y = \text{vm}$. Standard errors and confidence intervals are computed under Neyman-orthogonal scoring with $K = 5$ folds and $R = 2$ repetitions. reproducible from `morie::estimate_irm()` as documented in appendix A.

estimand	$\hat{\tau}$	se	t	95% CI
$\alpha\theta\epsilon$	0.1605	0.00628	25.54	[0.1481, 0.1728]
$\alpha\theta\theta\epsilon$	0.1557	0.00606	25.69	[0.1438, 0.1676]

3.1. Per-year δm estimates: rising trend

The same specification fit on each fiscal year separately produces the per-year estimates in Table 5. The point estimates rise monotonically from 1993 to 1998, and the per-year 95% confidence intervals do not nest the prior year for the $\alpha\theta\epsilon$ branch from 1993 to 1998.

Table 5: Per-fiscal-year iron-δm estimates of τ_{ATE} and τ_{ATTE} . The point estimates rise monotonically. reproducible from `morie::estimate_irm()` as documented in appendix A.

year	estimand	$\hat{\tau}$	se	95% CI
1993	$\alpha\theta\epsilon$	0.1342	0.00986	[0.1149, 0.1535]
1993	$\alpha\theta\theta\epsilon$	0.1272	0.01034	[0.1070, 0.1475]
1994	$\alpha\theta\epsilon$	0.1591	0.01167	[0.1362, 0.1820]
1994	$\alpha\theta\theta\epsilon$	0.1550	0.01141	[0.1326, 0.1774]
1998	$\alpha\theta\epsilon$	0.1737	0.01030	[0.1535, 0.1939]
1998	$\alpha\theta\theta\epsilon$	0.1704	0.00966	[0.1514, 0.1893]

3.3. multi-way clustered standard errors

To assess sensitivity to the within-individual and within-region correlation structure, the iron-δm estimator is refit under three clustering schemes (no clustering, cluster on identifier, cluster jointly on identifier and region) for each of five regional contrasts (Toronto-vs-rest, eastern-vs-rest, western-vs-rest, central-vs-rest, northern-vs-rest) and both estimands ($\alpha\theta\epsilon$, $\alpha\theta\theta\epsilon$), producing thirty estimator-clustering-region combinations.

Table 6: multi-way regional clustering robustness: range of $\hat{\tau}$ and se across the 30-cell grid (5 region contrasts $\times$ 3 clustering schemes $\times$ 2 estimands). all combinations satisfy $p < 10^{-24}$. reproducible from `morie::estimate_irm(cluster_by = ...)`.

Quantity	range
$\hat{\tau}$ across the 30 cells	[0.1932, 0.2013]
se across the 30 cells	[0.00346, 0.01923]
p -value (min)	$< 10^{-24}$

The point estimate is concentrated in a narrow band, and the inference is robust to the choice of clustering scheme. The min and max in Table 6 attach to specific cells: the minimum $\hat{\tau} =$

0.1932 corresponds to the `eastern-cluster_id=000` specification, and the maximum $\hat{\tau} = 0.2013$ corresponds to the `toronto-cluster_id+cluster_region=000` specification.

e.g. matched-sample Poisson gamma sensitivity

a 1:1 nearest-neighbour propensity-score match on (age, gender, year) followed by a Poisson generalised linear mixed model $vm \sim D + \text{age} + \text{gender} + \text{year} + (1 \mid \text{region})$ recovers an incidence-rate-ratio interpretation of the same treatment effect. The matched-sample sensitivity check confirms the direction and order of magnitude of the `mrm-δmλ` estimate: the matched-sample gamma and the `mrm-δmλ` are decision-equivalent in the sense that both reject the null and yield point estimates in the same $[0.13, 0.20]$ band. The exact gamma rate-ratio numbers are reproducible from `morie::run_treatment_effects_analysis(method = 'glmm_matched')` (appendix a), they are not central to the identification story and are deferred to the appendix tables.

e.e. identification robustness summary

across the pooled, per-year, and 36-cell clustering grid, the `mrm-δmλ` point estimate of the alert-complexity-to-placement-volatility effect lies in $[0.1272, 0.2013]$, with the matched-sample Poisson gamma falling in the same band on the matched subsample. The direction is unambiguous (positive), the size is order-of-magnitude stable, and the per-year trend is monotone rising over `1983--1988`.

d. mandela rules classification on 2017

The mandela rules ([united nations general assembly 1986](#)) are the united nations standard minimum rules for the treatment of prisoners (`artef/36/17`). rule 44 defines solitary confinement as confinement for 11 or more hours per day without meaningful human contact, and rule 43 prohibits the prolonged version of this practice (defined as exceeding fifteen consecutive days) as torture or other cruel, inhuman, or degrading treatment.

d.i. federal operationalisation

[prott and doob \(1983\)](#) jointly operationalise rules 43 and 44 on federal structured intervention unit person-stays:

$$\text{Class}(d, h, m) = \begin{cases} \text{Torture} & \text{if } d \geq 16, h \leq 2, m = 100, \\ \text{Solitary} & \text{if } d \leq 15, h \leq 2, m = 100, \\ \text{Allother} & \text{otherwise,} \end{cases} \quad (4)$$

where d is days in fiv, h is the average hours-out-of-cell per day, and $m \in [0, 100]$ is the percentage of days on which the inmate did not receive the legally-required four hours out-of-cell. across $N = 1,960$ federal fiv person-stays (november 1989 to september 1994), the classifier yields 28.4% solitary, 9.9% torture, and 61.7% all other ([prott and doob 1983](#)).

d.ii. Provincial operationalisation

the `oif` record does not expose hours-out-of-cell, so the `mrm` classifier applies a duration-only proxy at the provincial level:

$$\text{Class}^*(d) = \begin{cases} \text{Torture}^* & \text{if } d \geq 16, \\ \text{Solitary}^* & \text{if } 1 \leq d \leq 15. \end{cases} \quad (\text{e})$$

The proxy is an upper bound on the true mandela classification: some placements with $d \geq 16$ may in fact provide more than two hours out-of-cell on average, in which case the formal mandela definition is not met. The provincial application reads the official ontario `oif` `cii` aggregate file (`cii_individuals_in_segregation_and_restrictive_confinement_aggregate_lengths.csv`), which publishes per-fiscal-year counts of individuals by aggregate within-year duration band. The `Torture*` numerator is the sum of the $\{16\text{--}20, 21\text{--}25, 26\text{--}30, > 30\}$ -day bands; the denominator is the union over all duration bands (i.e., all individuals appearing in `oif` that fiscal year).

The provincial classifier is exposed in `morie` as `mrm.classify_mandela(cii_path, source = 'cii_aggregate')`.

D.3. cross-jurisdiction comparison

Table [7](#) reports the mandela classification for the federal structured intervention unit regime ([Jprot and Doob 1993](#)) alongside the provincial `oif` segregation and broader restrictive confinement views, by fiscal year. The provincial segregation proportion classified as `Torture*` rises from 12.5% in fiscal 1993 to 20.6% in fiscal 1998, a peak gap of +10.7 percentage points relative to the federal 9.9% that [Jprot and Doob \(1993\)](#) report. Under the broader restrictive-confinement view (`oif` `cii` `numberindividuals_restrictiveconfinement` column), the provincial `Torture*` rate reaches 40.9% in fiscal 1998.

Table [7](#): cross-jurisdiction mandela classification. Federal: person-stays, full operationalisation ([7](#)). Provincial: individuals, duration-only proxy ([e](#)). Source: [Jprot and Doob 1993](#) (federal) and the official ontario `oif` `cii` aggregate (provincial). All provincial numbers reproducible via `morie::mrm.classify_mandela(source = 'cii_aggregate')`.

Source	Period	Unit	Solitary %	Torture %	N
Federal JUS	1993--1998	Person-stays	28.4	9.9	1,960
Provincial seg. (<code>oif</code>)	1993	Individuals	87.5	12.5	12,647
Provincial seg. (<code>oif</code>)	1994	Individuals	83.5	16.5	10,881
Provincial seg. (<code>oif</code>)	1998	Individuals	79.4	20.6	9,608
Provincial rc (<code>oif</code>)	1993	Individuals	68.5	31.5	20,781
Provincial rc (<code>oif</code>)	1994	Individuals	64.0	36.0	19,641
Provincial rc (<code>oif</code>)	1998	Individuals	59.1	40.9	25,045

D.4. caveats

The federal--provincial comparison in Table [7](#) is directional rather than apples-to-apples. The federal numbers count person-stays and use the full out-of-cell-hour operationalisation of ([7](#)); the provincial numbers count individuals (which integrate across multiple placements within

a fiscal year) and use the duration-only proxy of (e). Two consequences follow. First, the provincial denominator is a different unit (individuals) from the federal denominator (person-stays). Second, the provincial numerator is an upper bound on the formal mandela classification, since some placements with $d \geq 16$ may have provided more than two hours out-of-cell on average. We nevertheless report the proxy alongside the federal figures because the magnitude of the gap is substantial even after these conservative adjustments are taken into account.

8. Statistical physics of crime (θ?) and institutional churn (ϕ?)

8.1. Per-category hawkes fits (markovian)

Table 8 reports the markovian (m) hawkes maximum-likelihood fits per θ? category, refitted with $u(u,i)$ -day jitter on the daily-resolution cc_{θ} .

Table 8: markovian hawkes (m) μ per category, sub-sampled to $n_{\text{fit}} = 1500$ events with $u(u,i)$ -day jitter. $\hat{\kappa}$ is the branching ratio. μp is the time-rescaling goodness-of-fit on the U_i residuals. Source: paper_hawkes_refit.json.

category	μ	$\hat{\kappa}$	μp	stationary?
assault	5062.8	0.972	0.889	yes
auto theft	4816.4	0.972	0.893	yes
bicycle theft	4767.2	0.969	0.623	yes
break ~ enter	5080.4	0.973	0.499	yes
homicide	2941.8	0.878	0.122	yes
robbery	5050.6	0.972	0.908	yes
shooting	4901.1	0.972	0.356	yes
theft from mv	5059.2	0.974	0.749	yes
theft over	5013.5	0.972	0.752	yes

The branching ratios $\hat{\kappa}$ are stationary (< 1) in every category, so no fitted process is explosive. Property crime ($\hat{\kappa} \approx 0.97$) sits very close to criticality. Homicide ($\hat{\kappa} = 0.878$) is the most sub-critical, in the sense that each homicide event triggers fewer follow-on offspring relative to baseline than any other category. The μp -values are all ≥ 0.12 , so no markovian fit is rejected by the time-rescaling test at conventional significance — but homicide ($p = 0.122$) is the closest to rejection, foreshadowing the much better non-markovian fit reported in Table 9.

8.1.1. markovian vs non-markovian hawkes

Table 9 contrasts the markovian classical hawkes against the kwan-chen-dunsmuir non-markovian weibull / sinusoidal-baseline alternative. The ΔAIC column is positive when the non-markovian model is preferred.

The non-markovian model is preferred in every single category, with ΔAIC ranging from 80.8 (homicide) to 133.6 (assault). Two patterns are worth noting. First, the non-markovian branching ratios $\hat{\eta}$ are uniformly lower than their markovian counterparts $\hat{\kappa}$ for the categories where the seasonal baseline carries a lot of signal: auto theft ($0.972 \rightarrow 0.814$), bicycle theft ($0.969 \rightarrow 0.667$), homicide ($0.878 \rightarrow 0.553$), robbery ($0.972 \rightarrow 0.764$), and theft over ($0.972 \rightarrow$

Table 9: markovian (exp / const) vs non-markovian (weibull / sin) hawkes per 07f category, both refitted under the same jitter scheme, $n_{\text{fit}} = 1500$ events. Positive $\Delta\text{AIC} = \text{aic}_{\text{mark}} - \text{aic}_{\text{nonmark}}$ favours the non-markovian model.

category	markovian		non-markovian (weibull / sin)		ΔAIC
	aic	$\hat{\kappa}$	aic	$\hat{\eta}$	
assault	5062.8	0.972	4929.1	0.965	133.6
auto theft	4816.4	0.972	4732.5	0.814	84.0
bicycle theft	4767.2	0.969	4633.6	0.667	133.5
break ~ enter	5080.4	0.973	4949.0	0.958	131.4
homicide	2941.8	0.878	2861.0	0.553	80.8
robbery	5050.6	0.972	4965.9	0.764	84.7
shooting	4901.1	0.972	4788.7	0.959	112.4
theft from mv	5059.2	0.974	4933.7	0.948	125.6
theft over	5013.5	0.972	4932.1	0.787	81.4

0.787) all see substantial reductions, as the sinusoidal baseline absorbs annual modulation that the markovian model was forced to attribute to self-excitation. Second, categories where $\hat{\eta}$ remains close to $\hat{\kappa}$ (assault $0.972 \rightarrow 0.965$, break and enter $0.973 \rightarrow 0.958$, shooting $0.972 \rightarrow 0.959$) are the ones with the strongest within-day or within-week self-excitation that the exponential kernel was already absorbing correctly.

The companion hawkes paper (private, /papers/hawkes-paper/) reports the full 6-way (kernel $\times$ baseline) comparison with all four kernel families and gives the kwan-chen-dunsmuir asymptotic theory in detail.

4.1.9 contiguous re-fit (v4.1.1 canonical for assault). The assault row of tables 9 and 9 above was computed on the legacy $n_{\text{fit}} = 1,500$ uniform-random sub-sample (a non-measure-preserving operation for a self-exciting process) and is retained for backward comparability with the v4.1.x analyses. The morie v4.1.1 release ships a gumba-jit'd full-sample fitter (described in the companion morie-py-paper and morie-r-paper) and the canonical post-v4.1.1 assault characterisation is the contiguous-year fit of 07f assault 4.1.9 ($n = 21,160$ events, $T = 365$ days, no sub-sampling, post-jitter clipping to $[0, T)$). That fit returns markovian $\hat{\eta} = 0.567, \hat{\beta} = 3.854/\text{day}$, w/ $p = 0.494$ (aic = $-130,068.4$)• exp/sinusoidal $\hat{\eta} = 0.510$, w/ $p = 0.332$ (aic = $-130,096.1$, $\Delta\text{aic} = 27.7$ over markovian)• and lmax/sinusoidal $\hat{\eta} = 0.513$, w/ $p = 0.298$ (aic = $-130,092.2$, 3.9 aic worse than exp/sinusoidal). The aic gain is attributable entirely to the sinusoidal baseline absorbing diurnal/weekly cyclicity rather than to a kernel-shape effect• the fat-tailed lmax kernel does not pay its extra-parameter cost on 07f assault 4.1.9. Full results are in data/manifest/outputs/paper_hawkes_assault_4.1.9_contiguous.json.

4.3. Lévy mobility, urban scaling, predator-prey

The Lévy column confirms the big 4.1.9 Brockmann et al. (4.1.9) prediction of heavy-tail offender mobility: every category lands in the heavy-tail regime $\alpha \in (1, 3)$. The Bettencourt column confirms the 4.1.9 Bettencourt et al. (4.1.9) super-linear violent-crime scaling within a single city's neighbourhoods for assault ($\hat{\beta} = 1.09$) and robbery ($\hat{\beta} = 1.12$)• ruhela also

Table i: Lévy-flight hill-slope exponent $\hat{\alpha}$ on inter-incident step lengths $\ell \geq 0.5$ km with $n_{\text{boot}} = 200$, Bettencourt per-ward super-linearity exponent $\hat{\beta}$ at year 2017, Lotka-volterra linearised cycle period T_{LV} . source: paper_findings_numbers.json.

category	$\hat{\alpha} \pm \sigma (n_{\text{tail}})$	$\hat{\beta} \pm \sigma (R^2)$	T_{LV} (yr)	flag
assault	1.331 ± 0.000 (28 307)	1.092 ± 0.112 (0.379)	160.7	--
auto theft	1.324 ± 0.000 (28 412)	1.130 ± 0.124 (0.347)	6 283.2	unidentified
bicycle theft	1.424 ± 0.001 (28 064)	0.812 ± 0.242 (0.071)	159.5	low R^2
break ~ enter	1.336 ± 0.001 (29 229)	0.854 ± 0.111 (0.274)	6 283.2	unidentified
fomicide	1.327 ± 0.002 (1 429)	0.114 ± 0.146 (0.011)	430.2	low R^2
robbery	1.332 ± 0.001 (26 685)	1.122 ± 0.162 (0.235)	198.6	--
fhooting	1.327 ± 0.001 (6 686)	0.588 ± 0.188 (0.074)	6 283.2	unidentified
theft from mv	1.337 ± 0.001 (28 998)	1.036 ± 0.098 (0.416)	6 283.2	unidentified
theft over	1.333 ± 0.001 (16 307)	1.488 ± 0.143 (0.410)	189.5	--

see the strongest super-linearity for theft over ($\hat{\beta} = 1.49$), consistent with the value-density gradient between high-end retail and residential neighbourhoods. The Lotka-volterra column should be read with care: $T_{\text{LV}} = 6\,283.2$ years $\approx 2\pi \cdot 10^3$ corresponds to fitted $\sqrt{\alpha\gamma} \rightarrow 0$ — i.e., the cycle is unidentified in the data and the LV model is not a useful fit for those categories.

x.u. Spatial autocorrelation and clustering

Table ii: global moran's I z -score, δ bscan cluster count at $\varepsilon = 0.3$ km, noise points, and largest-cluster size. all moran z -scores significant at $p < 10^{-3}$. source: paper_findings_numbers.json.

category	moran z	δ bscan n_c	noise	largest
assault	114.5***	179	5 301	10 177
auto theft	34.8***	271	7 723	1 914
bicycle theft	224.8***	53	4 164	22 528
break ~ enter	73.6***	185	8 059	10 841
fomicide	75.2***	1	1 499	32
robbery	94.9***	192	5 088	7 654
fhooting	96.5***	61	3 603	533
theft from mv	38.5***	199	8 250	9 734
theft over	49.4***	100	8 441	3 629

The bicycle theft mega-cluster ($n = 22\,528$, more than half of the category's events) localises the downtown commuter corridor — spatially the most concentrated category in the catalogue. The fomicide row ($n_c = 1$, largest cluster = 32) reflects that homicide is too rare to densify under δ bscan at $\varepsilon = 0.3$ km.

x.v. inspection-game phase diagram

The fieling-fzalnoki-erc replicator dynamics on $(T, \gamma) \in [0.05, 1.8] \times [0.05, 1.2]$ (a 20×20 grid) yields a steady-state defector-frequency mean of $\bar{x}_P^* = 0.262$, ranging from 0.000 in the cooperation-dominant regime to 0.955 in the predator-dominant regime. The analytical phase boundary $T^* \approx 1 + \gamma/2$ is recovered numerically: cooperation collapses sharply for $T > T^*$, and a mixed predator-cooperator corridor lies along the boundary, in agreement with [fieling](#)

et al. (2019), Jzolkani and Perc (2019). read substantively: an inspection regime with cost γ can sustain cooperation against predation temptation up to $T \lesssim 1 + \gamma/2$, beyond that cost-benefit margin the defector frequency rises monotonically.

3.1. reaction-diffusion: data-seeded vs Turing demo

The data-seeded β - δ solved on the cos-corrected Toronto grid produces a near-constant steady-state spike count of 109–118 across all nine θ categories ($\eta = 0.05$, $\omega = 0.3$, $\theta = 1.5$, $D = 0.1$, $\gamma = 0.05$, $dt = 0.04$, 800 steps on a 90×60 grid). empirical δ - β cluster counts on the same nine categories range from 1 (homicides, too sparse to densify at $\varepsilon = 0.3$ km) to 271 (auto theft). Because the β - δ steady-state spike count is a property of the β - δ parameters and grid resolution rather than of the underlying point process, the β -vs- δ spike-count comparison is largely a sanity check that the β - δ is in its spike-emitting regime, not a quantitative agreement. Only theft over (β 116 vs δ 100, deviation 16%) lands within a tight per-category tolerance. The clean-grid periodic Turing-regime simulation reproduces the canonical hexagonal hot-spot lattice from (Jhori et al. 2010, fig. 4), confirming that the implementation captures the qualitative phenomenology.

3.2. bulldorff space-time scan

For each category we ran a monte-carlo space-time bulldorff scan (bulldorff 1998) over $r \in \{1, 2, 3, 5, 8\}$ km candidate radii and 4-year time windows. The implementation is `morie.mrm_tps.kulldorff_scan()`. The candidate centres are sub-sampled from the event population, and the null distribution is formed by 199 permutations of the event timestamps. Table 1A reports the top cluster per category. Five of seven categories reject the no-clustering null at the $p = 0.005$ resolution of the permutation test (the floor for a 199-permutation test). Homicides and theft over are significant at $p = 0.040$, in both cases driven by lower event density that makes the cluster envelope smaller and the permutation distribution wider.

Table 1A: bulldorff top cluster per category, $r \in \{1, 2, 3, 5, 8\}$ km, 4-year window, 199 mc permutations on $n = 30,000$ random events per category (homicides and theft over are at full n). $\log L$ is the bulldorff Poisson log-likelihood-ratio, rr is the observed / expected ratio.

category	$\log L$	n_{obs}	rr	p	r (km)	window
assault	22.5	442	5.00	0.005	3	4444 -- 4444
auto theft	21.3	816	5.42	0.005	5	4440 -- 4444
break ~ enter	41.5	3490	3.23	0.005	5	4445 -- 4445
robbery	13.9	1168	4.85	0.005	5	4443 -- 4445
theft from mv	73.7	1585	6.72	0.005	2	4443 -- 4445
theft over	4.9	303	3.38	0.040	5	4443 -- 4445
homicides	5.0	12	3.51	0.040	1	4445 -- 4445

The theft-from-mv cluster ($\log L = 73.7$, $rr = 6.7$) is the strongest finding by likelihood and indicates a sustained 4443–4445 downtown corridor concentration. Cluster centres (43.5° – 43.8° N, -79.2° to -79.6° W) align with downtown Toronto, North York and Scarborough corridors, consistent with high-traffic urban networks. The cluster artefacts are written to `data/manifest/outputs/tps_spatial_advanced/kulldorff_topi.json`.

3.6. yoffmanian institutional churn (o0if)

yoffman’s i0bi asylums yoffman (i0bi) characterised “total institutions” as engines of mortification and churn: a small minority of inmates cycle repeatedly through the system, the institution’s classification machinery imposes identity-stripping moments on everyone, and inter-institutional transfers wash out personal trajectories. ruhela tests the structural predictions on o0if:

heavy-tailed repeat placement (verified against b09). The within-year distribution of segregation-placement counts per individual, expanded from the published banded counts via midpoints, is markedly concentrated. across pooled fiscal years 1983–1988 ($n = 33,136$ individuals contributing 153,067 placements) the yini coefficient is $G = 0.585$, and the top 5 of placed persons account for 25.9% of all placements — the signature heavy-tailed concentration yoffman predicted of total institutions. Per-year yinis rise monotonically over the panel ($0.542 \rightarrow 0.596 \rightarrow 0.616$), indicating intensifying placement inequality. The fill-m4e Pareto exponent at $x_{\min} = 1$ is $\hat{\alpha}_{\text{churn}} = 2.08$ pooled (1.11 / 1.10 / 1.10 per year), increasing to ~ 3.2 when $x_{\min}$ is pushed to ≥ 8 ; we adopt $x_{\min} = 1$ as the most-conservative reading since none of the bands below $x = 8$ visibly deviate from a straight tail on a log-log plot. The m4-thesis prose reports $\hat{\alpha} = 1.62$, but we are unable to reproduce that number from b09 at any $x_{\min}$; since the present paper is the verified record, we report $\hat{\alpha} = 2.08$.

cohort scope of the concentration result. The fact that the top 5 concentration is computed within fiscal year — rather than across the entire post-release career of an individual — is a strength of the within-year aggregate publication b09, not a limitation: o0if re-assigns uniqueindividual_id each fiscal year, so cross-year longitudinal tracking is impossible at the available unit of analysis. we accordingly omit the m4-thesis raplan-meier time-to-readmission claim from this paper: that quantity requires a stable cross-year identifier that the public release does not provide. within-year repeat-placement concentration remains valid, and is precisely what b09 is designed to capture.

mortification co-occurrence (verified against b0i). The three o0if b0i alert flags (mentalhealth_alert, fuiciderisk_alert, fuicidewatch_alert) co-occur within placement records to a degree well above independence. across all $n = 82,001$ b0i placements, the marginal yes-rates are 46.5%, 27.4% and 14.4%; any-of-three is flagged on 56.9% of placements and all-three simultaneously on 8.2%. The pairwise gramér’s V between mentalhealth_alert and fuiciderisk_alert is $V = 0.189$ ($\chi^2(1) = 2914$, $p < 10^{-300}$). The fuiciderisk $\times$ fuicidewatch association is much higher ($V = 0.668$) because watch is operationally a subset-escalation of risk in custody workflow. we adopt $V = 0.189$ as the substantive mortification co-occurrence figure, in agreement with the m4-thesis.

within-region locality (corrected interpretation of region churn). a 5×5 contingency table of region_at0imeofplacement $\times$ region_mostrecentplacement produces $\chi^2(16) = 285,917$ with $p < 10^{-3}$ (numerically zero) and gramér’s $V = 0.934$. The m4-thesis describes this as evidence of “routinised regional flows”; the verified table reverses the substantive interpretation. 95.0% of placements remain within the same region (77,866/82,001 on the diagonal) and only 5.0% involve a region change. what the χ^2 rejects is independence — but the deviation

from independence is the overwhelming diagonal (within-region locality), not cross-region mixing. ontario provincial segregation/rc placement is therefore best characterised as locality-preserving rather than as a regional-churn process: individuals are placed in, and tend to remain in, the region of their initial intake, with cross-region transfer restricted to a small minority of placements.

8.9. stochastic forecasting (farma / langevin / fokker-planck)

across all nine incident categories the `farma` fits absorb $> 80\%$ of monthly seasonality (verified by `acf` on residuals). `langevin` `ou` fits per category recover $\theta \in [0.07, 1.03]$ — slow mean-reversion at the low end (property crime), faster mean reversion ($\theta > 1$) for the most volatile categories — and $\sigma \in [0.5, 19.7]$, with several rare-event categories at $\sigma < 1$ indicating very low residual noise. the published `fokker-planck` manifest (`data/manifest/outputs/tps_stochastic/fokker_planck_*.json`) stores $(\theta, \mu, \sigma, \text{stationary_peak})$ per category together with a path to the rendered `KDE-v`s-stationary overlay `figs`. the overlays show close agreement between the fitted stationary density and the empirical `KDE` for the majority of categories, with bicycle theft visibly deviating because of strong annual seasonality that an `ou` model cannot represent.

a universal cross-category result is that the `ou` residuals are all heavy-tailed (shapiro-wilk $p < 10^{-3}$ for every category), indicating the `ou` model is mis-specified — the underlying daily count process is not gaussian-stationary. this is consistent with the hawkes self-exciting structure recovered in §8.1: a triggered hawkes process produces non-gaussian residuals when forced into an `ou` lens.

8.10. crime severity index — robustness check

the statistics canada crime severity index wallace et al. (1999) weights each criminal code offence by the product of average sentence length and proportion of offenders incarcerated, then sums per-capita. `csi` corrects for the volume / severity confound that haunts raw incident counts: a city where homicides drop but minor thefts rise will see falling `csi` even if absolute crime counts are flat.

table 13 reports `mpre`-computed toronto `csi` for 1997–1998 from the `morie.tps.csi` module, rebased to 1997 = 100. both the total and violent variants are shown. source weights: `statscan` catalogue 38-19-0199-01 (1993 update)* population: `statscan` 19-19-0000-01.

substantive read. toronto’s `csi` traces an unmistakable `U`-arc: a steady rise from 100 in 1997 to a 109 peak in 1999, a sharp drop to 91 in 2000 and a further floor at 89.0 in 2001, and a recovery to 107 in 2003 and 109 (violent) in 2004. the 1998 figure is for a partial calendar year and should not be read as a downturn. the violent `csi` tracks the total very closely, indicating the `U`-trough was driven by both violent and property categories — not a compositional shift.

cross-check against `statscan` published values. `statscan`’s published toronto `csi` for 2003 is approximately 60.4 (total) and 71.8 (violent) on its national 1990-baseline scale. the `mpre` module reports values at a different scale (raw weighted sum per 100k) because it does not have access to the 1990 national-baseline normalisation factor* rebasing locally to 1997 = 100

Table 13: Toronto crime severity index, $\Phi_{Y_{LHIQ-LHL\mathbb{E}}}$, $\eta_{\text{core-computed}}$ (morie.tps.csi.csi_per_year on the 9 θ_{f} open-data feeds, rebased so $LHIQ = iuu$). Total $\mathbb{C}_I$ weights all nine θ_{f} categories: violent $\mathbb{C}_I$ zeros out non-violent ($\beta^{\sim}\mathbb{E}$, auto theft, bicycle theft, theft from mv, theft over).

year	total $\mathbb{C}_I$ ($LHIQ=iuu$)	violent $\mathbb{C}_I$ ($LHIQ=iuu$)
$LHIQ$	100.0	100.0
$LHI\mathbb{E}$	99.5	100.7
$LHI\mathbb{D}$	103.2	107.3
$LHI\mathbb{Y}$	105.6	109.0
$LHI\mathbb{O}$	107.8	109.4
$LHI\mathbb{A}$	109.0 (peak)	108.4
$LHL\mathbb{H}$	92.0 (covid)	90.8 (covid)
$LHL\mathbb{I}$	86.6 (trough)	87.6 (trough)
$LHL\mathbb{A}$	96.5	97.3
$LHL\mathbb{Z}$	107.1	105.8
$LHLQ$	106.8	109.0
$LHL\mathbb{E}$	90.3 (partial)	91.7 (partial)

(as in Table 13) gives a clean within-Toronto trend that does not require the national baseline. For absolute comparison to other Canadian cities, use the `statscan`-published values directly.

$\omega_{\text{f}} \times \mathbb{C}_I$ $\alpha\mathbb{E}$ robustness overlay. we re-fit the §6 rm- $\delta\eta_A$ pipeline on the ω_{f} aui restrictive-confinement file with the matching year's θ_{f} -derived $\mathbb{C}_I$ value appended as an additional confounder, to test whether the headline $\alpha\mathbb{E}$ absorbs once we condition on the prevailing severity of the wider crime environment. The result is written to `data/manifest/outputs/otis_full/.../aui_csi/aui_with_csi_context.json` and gives a pooled $\hat{\tau}_{\alpha\mathbb{E}} = 0.213$ (cluster-robust $\text{se } 0.012$, $n = 14,520$, 95% $\mathbb{C}_I$ [0.189, 0.236]) with $\hat{\tau}_{\alpha\mathbb{O}\mathbb{E}} = 0.213$ and $\hat{\tau}_{\alpha\mathbb{O}\mathbb{C}} = 0.212$. The $\alpha\mathbb{E}$ is stable to within ≤ 0.002 of the $\mathbb{C}_I$ -blind estimate, so the severity of the surrounding crime environment is not a substantial confounder of the alert-complexity $\rightarrow$ placement-volatility relationship — the $\delta\eta_A$ ensemble already absorbs it through the demographic, regional and seasonal controls.

x.ii. extended spatial diagnostics

beyond global moran's I (Table 11), ruhela computed $L_{\text{f}}\mathbb{A}$ local indicators, yetis- $\text{ord } G_i^*$ z -scores, ripley's K , and per-year polygon moran's I from the `neighbourhoodcrimerates` `yeojf $\omega_{\text{f}}$` .

$L_{\text{f}}\mathbb{A}$ quadrants (verified). a local moran's I_i decomposition of the per-ward assault $LHLQ$ surface ($n = 158$ `fi $\omega_{\text{f}}$` polygons, $k = 6$ -nearest-neighbour weight matrix, 199 permutations) gives a global moran's $I = 0.270$ with quadrants `fi` 19.0%, `fi $\mathbb{A}$` 12.7%, `fi` 16.5%, `fi $\mathbb{A}$` 51.9%. among the $n_{\text{sig}} = 16$ wards significant at $p \leq 0.05$, the split is `fi` 68.8% (11/16), `fi $\mathbb{A}$` 31.2% (5/16): no `fi` or `fi $\mathbb{A}$` ward clears the 199-perm significance floor. The `fi $\mathbb{A}$` preponderance (52% of all wards) is the largest single group, and the significant cluster of 11 `fi` wards is the downtown core --- a bipolar structure where the substantive high-density signal is concentrated in a small core while the broader spatial pattern is dominated by quiet residential wards.

Per-year polygon moran's I , 1997--1999. Global moran's I on the per-ward assault counts is 0.376 in 1997, peaks at 0.437 in 1998, and declines monotonically to 0.270 in 1999. The interpretation is that assault has become less spatially concentrated over the panel: 2017--2019 saw a tighter hotspot regime, 2020 already softens during the pandemic, and 2022--2024 runs at the lowest values in the panel. output: data/manifest/outputs/tps_spatial_advanced/polygon_moran_per_year_assault.json.

getis-ord G_i^* (verified). $|G_i^*| > 1.96$ at 6--27 wards per category (most categories 16--19, the lowest is auto theft at 6 and the highest is homicides at 27, where the test is sensitive to a small number of high-rate wards relative to a quiet base). The strongest hot-spot z -score in our scan is $\hat{G}^* = 8.44$ on bicycle theft (downtown commuter corridor), shooting is at 6.83, assault at 6.76, and robbery at 6.23.

ripley's K . All nine categories exhibit $K(r) > \pi r^2$ at all distance thresholds 0.1--0.5 km, i.e., second-order clustering significantly above cr (complete spatial randomness). Property-crime $K(r)/\pi r^2$ ratios exceed violent-crime ratios at small radii, consistent with repeat-victimisation theory.

Pettitt change-point. On Toronto assault yearly counts 1997--1999, Pettitt's U change-point is detected at 1998 ($p < 0.01$), corresponding to the covid-19 lockdown regime change.

e. reproducibility: morie code for every reported result

every estimate in sections 2 and 3 is produced by a single morie call. The r-side calls are documented in appendix 4, the Python-side equivalents follow the parallel 4.1 documented in ruhela (1999b). A reader who installs morie install.packages('morie') (or pip install morie) and ingests the same open data through morie::mrm_otis_load() and morie::mrm_tps_load() recovers every reported number identically up to floating-point precision and learner random-seed choice.

f. Limitations

f.1. identifier reset prevents cross-year trajectories

The 2017 identifier-reset structure (section 1.1) prevents any analysis that would require tracking an individual across fiscal years. The per-year mrm-otid estimates of section 2.1 are therefore independent samples, not a longitudinal panel, the monotone rising trend is a population-level pattern, not a within-individual trajectory.

f.1.1. cross-file id joining is invalid

The same identifier across different b-series files of the same fiscal year does not denote the same person. Any analysis that joins bui to aui, c-tables, or d-tables on uniqueindividual_id

would semi-randomly pair distinct people. The morie bñ pipeline operates on bñ alone for the per-individual analysis of section 8.

9.3. identification assumptions

The irm-ð estimate of section 8 is identified under conditional exchangeability conditional on (age, gender, region, year). This is a strong assumption: unobserved clinical-severity covariates that drive both the alert-flag pattern and the regional-transfer logistics would violate it. The matched-sample Poisson glm sensitivity (section 8.4) tests robustness to the parametric outcome model: it does not test robustness to unobserved confounding.

10. Discussion

The empirical pattern is direction-stable across estimators (irm-ð, matched Poisson glm), clustering schemes (none, identifier, identifier $\times$ region), regional contrasts, and fiscal years. within the 36-cell clustering grid the point estimate is confined to [0.1932, 0.2013] with the per-year breakdown showing a monotone rising trend from 0.1342 in 1993 to 0.1737 in 1998. The matched Poisson glm sensitivity falls in the same order-of-magnitude band. The eleven statistical-physics-of-crime and institutional-churn result sections in section 4 document the broader empirical record on 01 and 01 bñ.

The findings should be interpreted as suggestive of a population-level pattern between alert complexity and placement volatility, conditional on the demographic and regional structure. The cross-year identifier-reset structure of 01 prevents the analysis from being read as a within-individual trajectory: the per-year breakdown should be read as three population snapshots rather than a panel.

11. conclusion

The mrm modules in morie produce a coherent empirical record on two open Canadian sources: 01 bñ (provincial restrictive confinement) and 01 (municipal-police events). The alert-complexity-to-placement-volatility effect is identified under the standard observational-inference assumptions: the eleven statistical-physics-of-crime and institutional-churn result sections in section 4 document fiawkes specifications, spatial diagnostics, yoffmanian institutional churn, and stochastic forecasting on the same data. every estimate in this paper is reproducible from the morie toolkit through the code in appendix 1.

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a. reproducibility code -- full morie commands for every reported result

The r-side code below is run on the `otf-rs` environment data loaded through `morie::mrm_otis_load(year = 1983:1988)`.

a.i. Pooled $\delta\eta\lambda$ (table 4)

```
r> library(morie)
r> df <- mrm_otis_load(year = 1983:1988)
r> res_pool <- estimate_irm(
+   y = 'vm',
+   d = 'ac'gea',
+   x = c('age', 'gender', 'region', 'year'),
+   data = df,
+   n_folds = 5, n_rep = 1)
r> print(res_pool) # expect  $\alpha\theta\epsilon = 1.1046$  (JE 1.111046)
```

a.ii. Per-year $\delta\eta\lambda$ (table 5)

```
r> res_by_year <- lapply(1983:1988, function(y) ||
+   estimate_irm(
+     y = 'vm', d = 'ac'gea',
+     x = c('age', 'gender', 'region'),
+     data = subset(df, year == y),
+     n_folds = 5, n_rep = 1)
+ #)
```

a.iii. multi-way clustering (table 6)

```
r> regions <- c('toronto', 'eastern', 'western', 'central', 'northern')
r> schemes <- list(
+   none = NULL,
+   id = 'id',
+   idrgn = c('id', 'region'))
r> res_all <- expand_grid(region = regions, scheme = names(schemes))
r> res_all_fit <- mapply(function(r, sname) ||
+   estimate_irm(
+     y = 'vm', d = 'ac'gea',
+     x = c('age', 'gender', 'year'),
+     data = subset(df, region == r | region != r),
+     cluster_by = schemes[[sname]])
+ #, res_all_region, res_all_scheme, SIMPLIFY = FALSE)
```

a.iv. matched-sample Poisson χ^2 test (section 5.4)

```
r> res`glmm ~- run`treatment`effects`analysis(
+   data = df, treatment = 'ac`geΛ', outcome = 'vm',
+   covariates = c('age', 'gender', 'year'),
+   cluster`by = 'region',
+   method = 'glmm`matched')
```

a.v. hawkes on $\theta\mathcal{I}\mathcal{J}$ (section ??)

```
r> tps ~- mrm`tps`load(category = 'assault', year = ΛΗΙΨ:ΛΗΛΘ)
r> hawkes`fits ~- mrm`hawkes`tps(
+   events = tps,
+   kernels = c('exponential', 'gamma', 'weibull', 'power`law'),
+   baselines = c('constant', 'sinusoidal'))
r> print(hawkes`fits`selection) ~ αΙΕ-ranked, βJ-tested
```

β. extended clustering schemes

Two specifications additional to those in section 3.3 are run by the morie estimate_irm() wrapper for completeness: (i) cluster on fiscal year alone, and (ii) include identifier as a covariate and cluster on year. The point estimates in both specifications fall within the [0.1932, 0.2013] band reported in Table D, confirming that the inference does not depend on the choice between identifier-level, region-level, or year-level clustering.

γ. hawkes per-category details

The per-category αΙΕ ranking, branching-ratio estimates, and kolmogorov--smirnov p -values for the time-rescaling residuals are produced by the morie call documented in appendix A above and are not reproduced in the print body of this paper to avoid duplication with ruhela (ΛΗΛΔα), where the per-category fits are reported in full.

references

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